

Example to show the working of Runge Kutta method: (Rachel Batra)

1 Solve $dy/dx = x$, where at $x = 0$, $y = 0$.

1) We solve the above differential eqn using ode2 method;

```
--> eqn : 'diff ( y , x ) = x ;  
sol : ode2 ( eqn , y , x ) ;  
fsol : ic1 ( sol , x = 0 , y = 0 ) ;  
wxplot2d ( rhs ( fsol ) , [ x , 0 , 2 ] ) $
```

2) Using desolve

```
--> kill ( all ) $  
  
--> eqn : diff ( y ( x ) , x ) = x ;  
sol : desolve ( eqn , y ( x ) ) ;  
fsol : ev ( sol , y ( 0 ) = 0 ) ;  
wxplot2d ( rhs ( fsol ) , [ x , 0 , 2 ] ) $
```

3) using Runge Kutta method

For one differential equation:

i). rk(ODE, VAR, INITIAL, DOMAIN)

It solves numerically one first-order ordinary differential equation where

ODE: we write the rhs of the first order differential equation

VAR: represents the dependent variable

INITIAL: the initial value of the dependent variable when independent variable is x0.

DOMAIN: the independent variable is specified with domain, which must be a list of four elements. eg [x, x0, xn, h]

here xn is the last value at which we approximate y(xn)

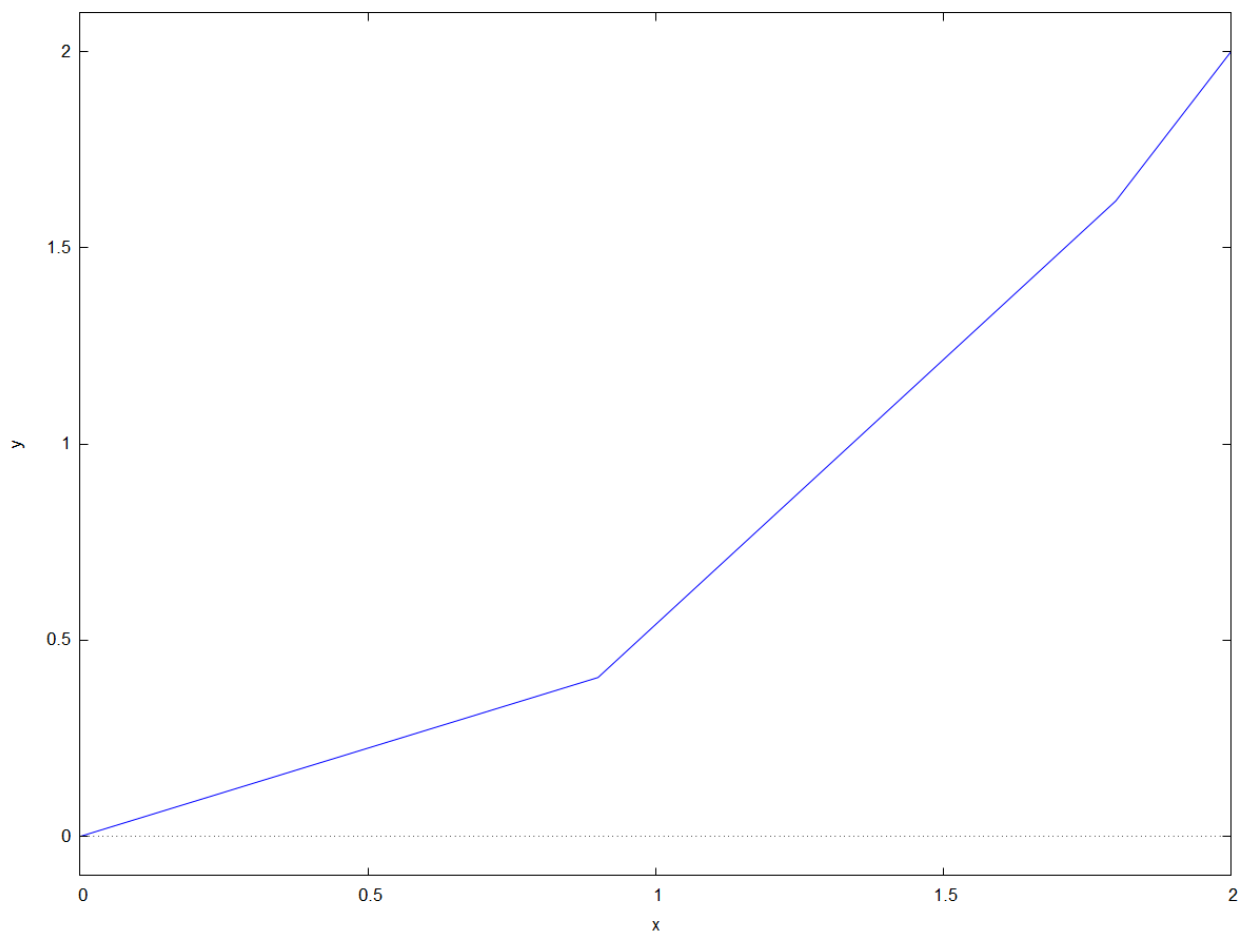
For a system of differential equations

ii). rk([ODE1,...,ODEm],[v1,...,vm],[init 1,...,init m], domain)

```
--> eqn : 'diff ( y , x ) = x ;  
fsol : rk ( rhs ( eqn ) , [ y ] , [ 0 ] , [ x , 0 , 2 , 0.9 ] ) ;  
wxplot2d ( [ discrete , fsol ] ) $
```

$$\frac{d}{dx}y = x$$

[[0.0 , 0.0] , [0.9 , 0.405] , [1.8 , 1.62] , [2.0 , 2.0]]



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Created with [wxMaxima](#).

The source of this Maxima session can be downloaded [here](#).